

Zhao Zhang

94, Brett Rd, Piscataway, NJ 08854 | zhao.zhang@rutgers.edu

CURRENT POSITION

Department of ECE, Rutgers University <i>Assistant Professor</i>	09/2023-present
--	-----------------

EDUCATION

University of Chicago <i>Ph.D Computer Science</i>	09/2009 – 06/2014
University of Chicago <i>M.S. Computer Science</i>	01/2007 – 12/2007
Beijing University of Posts and Telecommunications <i>B.E. Software Engineering</i>	09/2002 – 06/2006

EXPERIENCE

Texas Advanced Computing Center <i>Research Scientist, Machine Learning Group Leader</i>	06/2022 - 08/2023 <i>Austin, TX</i>
Texas Advanced Computing Center <i>Research Associate, Data Intensive Computing Group</i>	07/2016 – 05/2022 <i>Austin, TX</i>
AMPlab and Berkeley Institute for Data Science, UC Berkeley <i>Postdoc Researcher and Data Science Fellow</i>	07/2014 – 05/2016 <i>Berkeley, CA</i>
Computation Institute, University of Chicago <i>Scientific Application Specialist</i>	01/2008 – 07/2009 <i>Chicago, IL</i>

ACTICE PROJECTS

-
- PI, NSF OAC-2340011 “Collaborative Research: Frameworks: hpcGPT: Enhancing Computing Center User Support with HPC-enriched Generative AI” (8/1/24-7/31/27)
 - PI, NSF OAC-2340011 “CAREER: Efficient and Scalable Large Foundational Model Training on Supercomputers for Science” (10/1/23-9/30/26)
 - PI, NSF OAC-2311766 “Collaborative Research: Frameworks: Diamond: Democratizing Large Neural Network Model Training for Science” (10/1/23-9/30/26)
 - PI, NSF CNS-2312689 “Collaborative Research: CSR: Medium: Fortuna: Characterizing and Harnessing Performance Variability in Accelerator-rich Clusters” (10/1/23-9/30/26)
 - PI, NSF OAC-2106661 “Collaborative Research: OAC Core: ScaDL: New Approaches to Scaling Deep Learning for Science Applications on Supercomputers” (10/1/21-9/30/24)
 - Rutgers PI, NSF OAC-2112606 “AI Institute for Intelligent CyberInfrastructure with Computational Learning in the Environment (ICICLE)” (11/1/21-10/31/26)

PAST PROJECTS

-
- PI, NSF OAC-2008388 “Collaborative Research: OAC Core: Small: Efficient and Policy-driven Burst Buffer Sharing” (10/1/20-9/30/22)
 - co-PI, NSF OAC-1931537 “Collaborative Research: Frameworks: Designing Next-Generation MPI Libraries for Emerging Dense GPU Systems” (11/1/19-10/31/22)
 - PI, UTA20-000807 “Implementation of Performance Optimization for Machine Learning Applications at Total” (07/01/2020-06/30/2023)

- [MLSys'25] Zheng, Mingkai and **Zhao Zhang**. "Radius: Range-based Gradient Sparsity for Large Foundation Model Pre-training." *to appear in MLSys'25*.
- [HPDC'25] Sun, Yu, Zhu Zhu, Cherish Mulpuru, Roberto Gioiosa, **Zhao Zhang** Bo Fang, Lishan Yang "FT2: First-Token-Inspired Online Fault Tolerance on Critical Layers for Generative Large Language Models." *to appear in HPDC'25*
- [ICLR'25] Mozaffari, Mohammad, Amir Yazdanbakhsh, **Zhao Zhang**, and Maryam Mehri Dehnavi. "SLoPe: Double-pruned sparse plus lazy low-rank adapter pretraining of LLMs." *to appear in ICLR'25*
- [NeurIPS'24] Mozaffari, Mohammad, Sikan Li, **Zhao Zhang**, and Maryam Mehri Dehnavi. "MKOR: Momentum-Enabled Kronecker-Factor-Based Optimizer Using Rank-1 Updates." *in Advances in Neural Information Processing Systems 36 (2024)*.
- [SC'23] Q. Ding, P. Zheng, S. Kudari, S. Venkataraman, **Z. Zhang**. "Mirage: Towards Low-interruption Services on Batch GPU Clusters with Reinforcement Learning." *In Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, pp. 1-13. 2023.
- [SC'23] E. Karrels, L. Huang, Y. Kan, I. Arora, Y. Wang, D. S. Katz, B. D. Gropp, **Z. Zhang**. "Fine-grained Policy-driven I/O Sharing for Burst Buffers." *In Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, pp. 1-13. 2023.
- [Euro-Par'23] Zhang, Bo, Philip E. Davis, Nicolas Morales, **Zhao Zhang**, Keita Teranishi, and Manish Parashar. "Optimizing Data Movement for GPU-Based In-Situ Workflow Using GPUDirect RDMA." *In European Conference on Parallel Processing*, pp. 323-338. Cham: Springer Nature Switzerland, 2023.
- [SC'21] J. G. Pauloski, Q. Huang, L. Huang, K. Chard, I. T. Foster, **Z. Zhang**. "KAISA: An Adaptive Second-order Optimizer Framework for Deep Neural Networks." *in Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis (SC)*, pp. 1-14. 2021.
- [ISC'21] Evans, Richard Todd, Matthew Cawood, Stephen Lien Harrell, Lei Huang, Si Liu, Chun-Yaung Lu, Amit Ruhela, Yinzhi Wang, and **Zhao Zhang**. "Optimizing GPU-Enhanced HPC System and Cloud Procurements for Scientific Workloads." *In International Conference on High Performance Computing*, pp. 313-331. Springer, Cham, 2021.
- [SC'20] J. G. Pauloski, **Z. Zhang**, L. Huang, W. Xu, I. T. Foster. "Convolutional Neural Network Training with Distributed K-FAC." *In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis (SC)*, pp. 1-12. IEEE, 2020.
- [IPDPS'20] **Z. Zhang**, L. Huang, J. G. Pauloski, I. T. Foster. "Efficient I/O for Neural Network Training with Compressed Data." *In 2020 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, pp. 409-418. IEEE, 2020.
- [CLUSTER'19] **Z. Zhang**, L. Huang, R. Huang, W. Xu, D. S. Katz. "Quantifying the Impact of Memory Errors in Deep Learning." *In Proceedings of 2019 IEEE International Conference on Cluster Computing (CLUSTER)*, p.1-12, 2019.
- [ICPP'18] Y. You, **Z. Zhang**, J. Demmel, K. Keutzer, C. Hsieh. "ImageNet Training in Minutes." *In Proceedings of the 47th International Conference on Parallel Processing*, p. 0. ACM, 2018. **Best Paper**
- [HPDC'17] **Z. Zhang**, E. Sparks, M. J. Franklin. "Diagnosing Machine Learning Pipelines with Fine-grained Lineage." *In Proceedings of the 26th international symposium on High-performance parallel and distributed computing (HPDC' 17)*
- [IPDPS'16] M. Turilli, F. Liu, **Z. Zhang**, A. Merzky, M. Wilde, J. Weissman, D. S. Katz, and S. Jha. "Integrating abstractions to enhance the execution of distributed applications." *In 2016 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, pp. 953-962. IEEE, 2016

- [CIDR'15] D. Crankshaw, P. Bailis, J. E. Gonzalez, H. Li, **Z. Zhang**, M. J. Franklin, A. Ghodsi, M. I. Jordan. "The Missing Piece in Complex Analytics: Low Latency, Scalable Model Management and Serving with Velox." *7th Biennial Conference on Innovative Data Systems Research (CIDR)*, 2015.
- [SIGMOD'15] F. A. Nothaft, M. Massie, T. Danford, **Z. Zhang**, U. Laserson, C. Yeksigian, J. Kottalam et al. "Rethinking data-intensive science using scalable analytics systems." *In Proceedings of the 2015 ACM SIGMOD International Conference on Management of Data*, pp. 631-646. ACM, 2015.
- [eScience'14] **Z. Zhang**, D. S. Katz. "Using Application Skeletons to Improve eScience Infrastructure." *In e-Science (e-Science), 2014 IEEE 10th International Conference on*, vol. 1, pp. 111-118. IEEE, 2014.
- [SC'13] **Z. Zhang**, D. S. Katz, T. Armstrong, J. Wozniak, I. Foster. "Parallelizing the Execution of Sequential Scripts." *In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis (Supercomputing)*, p. 31. ACM, 2013.
- [IPDPS'13] T. Li., X. Zhou, K. Brandstatter, D. Zhao, K. Wang, A. Rajendran, **Z. Zhang**, and I. Raicu. "ZHT: A light-weight reliable persistent dynamic scalable zero-hop distributed hash table." *In 2013 IEEE 27th International Symposium on Parallel and Distributed Processing*, pp. 775-787. IEEE, 2013.
- [HPDC'12] **Z. Zhang**, D. S. Katz, M. Wilde, J. Wozniak, I. Foster. "MTC Envelope: Defining the Capability of Large Scale Computers in the Context of Parallel Scripting Applications." *In Proceedings of the 22nd international symposium on High-performance parallel and distributed computing (HPDC)*, pp. 37-48. ACM, 2013.
- [SC'12] **Z. Zhang**, D. S. Katz, J. Wozniak, A. Espinosa, I. Foster. "Design and Analysis of Data Management in Scalable Parallel Scripting." *In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis (Supercomputing)*, p. 85. IEEE Computer Society Press, 2012.
- [SC'08] I. Raicu, **Z. Zhang**, M. Wilde, I. Foster, P. Beckman, K. Iskra, B. Clifford. "Toward Loosely Coupled Programming on Petascale Systems." *In Proceedings of the 2008 ACM/IEEE conference on Supercomputing*, p. 22. IEEE Press, 2008.

JOURNAL PUBLICATIONS

- [Nature Methods'24] Ahdriz, Gustaf, Nazim Bouatta, Christina Floristean, Sachin Kadyan, Qinghui Xia, William Gerecke, Timothy J. O'Donnell et al. "OpenFold: Retraining AlphaFold2 yields new insights into its learning mechanisms and capacity for generalization." *Nature Methods (2024)*: 1-11.
- [TPDS'22] J. G. Pauloski, L. Huang, W. Xu, I. T. Foster, **Z. Zhang**. "Deep Neural Network Training with Distributed K-FAC." *in IEEE Transactions on Parallel and Distributed Systems*, doi: 10.1109/TPDS.2022.3161187.
- [Nature Methods'21] Fang, Linjing, Fred Monroe, ..., **Z. Zhang**, et al. "Deep learning-based point-scanning super-resolution imaging." *Nature Methods (2021)*: 1-11.
- [TPDS'19] Y. You, **Z. Zhang**, J. Demmel, K. Keutzer, C. Hsieh. "Fast Deep Neural Network Training on Distributed Systems and Cloud TPUs." *IEEE Transactions on Parallel and Distributed Systems* 30, no. 11 (2019): 2449-2462.
- [TBD'16] **Z. Zhang**, K. Barbary, F. A. Nothaft, E. Sparks, O. Zahn, M. J. Franklin, D. A. Patterson, S. Perlmutter. "Kira: Processing Astronomy Imagery Using Big Data Technology." *IEEE Transactions on Big Data*, Volume: 3. 2016
- [CCPE'16] T. Li, X. Zhou, K. Wang, D. Zhao, I. Sadooghi, **Z. Zhang**, and I. Raicu. "A convergence of key-value storage systems from clouds to supercomputers." *Concurrency and Computation: Practice & Experience* 28, no. 1 (2016): 44-69.

- [FGCS'15] D. S. Katz, A. Merzky, **Z. Zhang**, S. Jha. "Application Skeletons: Construction and Use in Science." *In Press, Future Generation Computer Systems (FGCS)*, 2015.
- [CLUSTER'10] I. Raicu, I. Foster, M. Wilde, **Z. Zhang**, K. Iskra, P. Beckman, Y. Zhao et al. "Middleware support for many-task computing." *Cluster Computing* 13, no. 3 (2010): 291-314.
- [COMPUTER'09] M. Wilde, I. Foster, K. Iskra, P. Beckman, **Z. Zhang**, A. Espinosa, M. Hategan, B. Clifford, and I. Raicu. "Parallel scripting for applications at the petascale and beyond." *Computer* 42, no. 11 (2009): 50-60.

PROFESSIONAL SERVICE

- **Co-lead of Model Pretraining Runtime Thrust** in the Trillion Parameter Consortium Project.
- **Co-lead of Bootcamp and Tutorial Committee** in the Cyberinfrastructure for AI for Science and Society, Rutgers.
- **Technical Program Committee Member**, eScience'25.
- **Technical Program Committee Member**, Machine Learning and HPC Track, SC'20, 21, 22, 24.
- **Technical Program Committee Member**, Architecture Track, IPDPS'24.
- **Review Board Member**, IEEE Transaction on Parallel and Distributed Computing, 2023
- **Technical Program Committee Member**, HPDC'23.
- **Technical Program Committee Member**, Architecture Track, ICPP'23.

TUTORIALS

- Tools and Best Practices for Distributed Deep Learning on Supercomputer, Tutorial, SC'18, SC'19, SC'20, SC'21
- Distributed Deep Learning, Tutorial, TACC Machine Learning Institute, 2018, 2019, 2020
- Introduction to ML/DL and its Applications in Natural Hazard, DesignSafe Bootcamp, 2020
- ML4GEO training for Petrobras, 2018
- Spark Internals, TACC Machine Learning Institute, 2017

STUDENTS

Current Ph.D Students

- Shuyuan Fan 01/24-present Ph.D student in ECE
- Jingxin Wang 01/24-present Ph.D student in ECE
- Mingkai Zheng 01/24-Present, Ph.D student in ECE
- Haotian Xie 09/24-present Ph.D student in ECE
- Kun Lin 09/24-present Ph.D student in ECE
- Molang Wu 09/24-present Ph.D student in ECE
- Fangru Linghu 01/25-present Ph.D student in ECE
- Daiheng Zhang 01/25-present Ph.D student in ECE

Current Undergraduate Students

- Michael Mogilevsky, 02/24 - present, B.S. student in ECE
- Hazem Zaky, 02/24 - present, B.S. student in ECE

Graduated Students

- Yuhong Kan, 01/22-05/23, M.S. in ECE (Software Engineer at Bloomberg)
- Yue Shangguan, 09/21-05/23, M.S. in ECE (Software Engineer at Bloomberg)
- Qiyang Ding, 09/21-05/23, M.S. in ECE (Hardware Engineer at Apple)
- Sikan Li, 06/21-05/22, M.S. in ECE (Research Engineer at TACC)
- Shreyas Kudari, 09/21-05/22, B.S. in ECE (Software Engineer at Meta)
- Ishank Arora, 09/20-05/22, M.S. in CS (Software Engineer at Apple)
- Qi Huang, 09/20-05/21, M.S. in CS (Software Engineer at Bloomberg)
- J. Gregory Pauloski, 05/19-08/20, now a Ph.D Student in CS at UChicago

REFERENCE

- **Niall Gaffney**
Director of Data Intensive Computing, Texas Advanced Computing Center
Email: ngaffney@tacc.utexas.edu
- **Michael J. Franklin**
Chair, Department of Computer Science, UChicago
Email: mjfranklin@cs.uchicago.edu
- **Saul Perlmutter**
Director, Berkeley Institute for Data Science, UC Berkeley
Franklin W. and Karen Weber Dabby Professor of Physics, UC Berkeley
Email: saul@lbl.gov
- **Ian T. Foster**
Director, Computation Institute, UChicago & ANL
Arthur Holly Compton Distinguished Service Professor of Computer Science, UChicago
Email: foster@anl.gov
- **Daniel S. Katz**
Assistant Director, Scientific Software and Applications, National Center for Supercomputing Applications (NCSA)
Email: d.katz@ieee.org